

INTERNSHIP REPORT
ON
SMART VIBRATION DETECTION GLOVES
(A PROJECT REPORT)

Submitted by

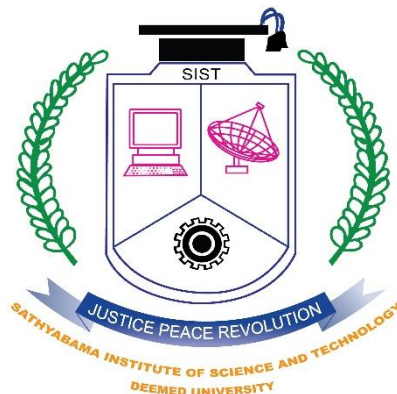
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ABSTRACT

In the automotive manufacturing industry, ensuring the proper installation of wiring harness locks and connectors is essential for maintaining product quality, safety, and production efficiency. Conventional inspection methods rely on manual verification, which can be time-consuming and prone to human error. To address this challenge, this project presents the development of an **Industrial Smart Glove for Zero False Detection of Harness Lock Installation** using embedded systems, sensor fusion, and vibration-based analysis. The system is designed to provide real-time verification of locking operations and assist operators in achieving consistent assembly quality.

The proposed smart glove uses a Blue Pill (STM32F103C8) microcontroller with MPU6050 and PVDF sensors to detect motion and locking impacts. A state-machine architecture and FFT analysis are used to process vibration signals. During calibration, vibration data from five correct lock installations are recorded and averaged to create a Golden Template. This template is stored in EEPROM and used for future lock validation.

During operation, the system captures vibration data after each locking impact and compares its frequency spectrum with the stored Golden Template. By reducing false detections caused by hand movements and external noise, the system improves assembly accuracy, minimizes rework, and enhances productivity. The project demonstrates the use of smart wearable technology and digital signal processing for real-time quality assurance in automotive manufacturing.

TABLE OF CONTENTS

CHAPTER 1 - Introduction

CHAPTER 2 -Problem Statement and Objective

CHAPTER 3 -Requirement Analysis

CHAPTER 4 –System Architecture

CHAPTER 5 – Application Logic and Feature Break Down

CHAPTER 6 - Conclusion

CHAPTER 1

INTRODUCTION

In the automotive manufacturing industry, the correct installation of wiring harness connectors and locking mechanisms is essential to ensure vehicle safety, reliability, and product quality. Traditionally, the verification of harness lock installation is performed through manual inspection or operator experience, which can be time-consuming and susceptible to human error. In high-volume production environments, even a small assembly mistake can lead to rework, increased production costs, and potential quality issues. Therefore, there is a growing need for intelligent systems that can assist operators in verifying assembly operations accurately and in real time.

This project focuses on the development of an Industrial Smart Glove for Zero False Detection of Harness Lock Installation. The proposed system combines embedded electronics, sensor fusion, and digital signal processing techniques to identify successful locking events. The smart glove is designed using a Blue Pill (STM32F103C8) microcontroller integrated with an MPU6050 motion sensor, a PVDF piezoelectric vibration sensor, and a WS2812B RGB LED indicator. By monitoring hand movement and vibration signals generated during locking operations, the system can distinguish valid lock installations from unwanted motions and environmental disturbances.

A state-machine-based approach is implemented to ensure reliable operation. The MPU6050 sensor detects hand motion, while the PVDF sensor captures impact vibrations produced during harness locking. The captured vibration signals are processed using the Fast Fourier Transform (FFT) algorithm to obtain their frequency characteristics. During calibration, vibration data from five correctly installed locks are recorded and averaged to create a Golden Template, which is stored in EEPROM memory. During real-time operation, newly acquired

vibration signatures are compared with this reference template to determine the quality of the lock installation.

The developed smart glove provides immediate visual feedback through LED indications, enabling operators to identify successful and defective lock installations instantly. By reducing false detections, minimizing inspection effort, and improving assembly accuracy, the proposed system contributes to enhanced productivity and quality assurance in automotive manufacturing. Furthermore, the project demonstrates the practical application of Industry 4.0 concepts, smart wearable technology, and intelligent sensing systems in modern industrial environments.

CHAPTER 2

PROBLEM STATEMENT AND OBJECTIVE

PROBLEM STATEMENT

The verification of wiring harness lock installation in automotive assembly lines is traditionally dependent on operator experience, manual inspection, or basic single-sensor detection methods. Earlier systems attempted to identify lock engagement using sound-based detection techniques; however, these approaches suffered from high false detection rates and poor reliability in industrial environments. The sound generated during a lock installation can be easily masked by surrounding machinery noise, tool impacts, worker movement, and other production activities, making accurate lock verification difficult.

Limitations of Existing Systems

High False Detection Rate:

Sound-based systems often misinterpret unrelated noises such as tool drops, metal contact, machine vibrations, or worker activities as valid lock events, resulting in frequent false alarms and reduced system reliability.

Lack of Accurate Lock Validation:

Existing methods focus only on detecting the audible "click" sound and cannot verify whether the lock has been properly engaged. As a result, defective or incomplete lock installations may still be classified as successful.

Environmental Interference:

Industrial assembly environments contain significant background noise and vibrations that affect sensor performance. Single-sensor systems are unable to distinguish actual locking events from external disturbances.

Absence of Real-Time Intelligent Analysis:

Traditional inspection methods do not analyze the unique vibration signature

generated during lock engagement. Without advanced signal processing, subtle differences between correct and incorrect installations remain undetected.

Limited Operator Feedback:

Most existing systems lack instant feedback mechanisms, forcing operators and quality inspectors to rely on manual verification, which increases inspection time and the possibility of human error.

Need for the Proposed System

To overcome these limitations, there is a need for an intelligent wearable system capable of accurately identifying successful harness lock installations while minimizing false detections. The proposed Smart Glove addresses these challenges by combining motion sensing, vibration analysis, sensor fusion, state-machine logic, and FFT-based signal processing. By analyzing the actual vibration signature of a lock installation rather than relying solely on sound, the system provides reliable real-time validation, immediate operator feedback, improved assembly accuracy, and enhanced quality assurance in automotive manufacturing environments.

OBJECTIVES OF THE PROJECT

Primary Objective

- To develop an Industrial Smart Glove capable of accurately validating wiring harness lock installations in real time using sensor fusion and vibration signature analysis, thereby achieving near-zero false detections in automotive assembly operations.

Specific Objectives

- To design and develop a wearable embedded system using the Blue Pill (STM32F103C8) microcontroller for intelligent harness lock verification.

- To integrate an MPU6050 motion sensor and a PVDF piezoelectric vibration sensor for capturing hand movement and lock-impact signals during assembly operations.
- To implement a Sensor Fusion Logic that combines motion data and vibration data to distinguish genuine lock installation events from random movements, accidental impacts, and environmental disturbances.
- To develop a State Machine Architecture consisting of Idle, Motion Detection, Impact Detection, and Analysis states for reliable event processing and decision making.
- To establish a robust Calibration Process by recording multiple correctly installed harness locks and generating a reference vibration profile.
- To create a Golden Template by averaging vibration signatures obtained from five successful lock installations and storing the template in EEPROM memory for future comparison.
- To implement Fast Fourier Transform (FFT) analysis for converting time-domain vibration signals into frequency-domain data and extracting unique locking characteristics.
- To compare real-time FFT results with the stored Golden Template and calculate a similarity score for lock validation.
- To provide instant visual feedback through WS2812B RGB LED indicators for successful and defective lock installations.
- To eliminate false detections associated with conventional sound-based or single-sensor detection systems by utilizing vibration-based frequency signature matching.
- To improve assembly accuracy, reduce quality defects, minimize rework, and enhance productivity in automotive manufacturing environments.

- To demonstrate the practical application of embedded systems, sensor fusion, calibration algorithms, digital signal processing, and Industry 4.0 technologies for real-time industrial quality assurance.

CHAPTER 3

REQUIREMENT ANALYSIS

Software Requirements

The software development and implementation of the Smart Glove system were carried out using the **Arduino IDE**, which provides a user-friendly platform for programming, compiling, and uploading firmware to the Blue Pill (STM32F103C8) microcontroller. The system software integrates sensor acquisition, signal processing, state-machine control, calibration routines, and real-time feedback mechanisms.

Software Tools

- Arduino IDE
- STM32 Board Package for Arduino
- Serial Monitor for debugging and calibration

Libraries Used

Library	Purpose
EEPROM.h	Stores and retrieves the Golden Template in non-volatile memory
arduinoFFT.h	Performs Fast Fourier Transform (FFT) analysis on vibration signals
Wire.h	Enables I2C communication between the microcontroller and MPU6050 sensor
MPU6050_light.h	Reads acceleration and motion data from the MPU6050 sensor

Adafruit_NeoPixel.h	Controls the WS2812B RGB LED for visual feedback
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Hardware Requirements

The Smart Glove system consists of sensors, processing units, and output devices that work together to detect and validate harness lock installations.

Hardware Components

Component	Specification	Purpose
Blue Pill Microcontroller	STM32F103C8	Main processing and control unit
MPU6050 Sensor	3-Axis Accelerometer & Gyroscope	Detects hand movement and motion
PVDF Piezoelectric Sensor	Vibration Sensor	Captures impact vibrations generated during lock installation
WS2812B RGB LED	Smart LED	Provides visual indication of lock validation results
EEPROM Memory	Internal MCU Memory	Stores Golden Template calibration data
USB Cable	Programming Interface	Uploads firmware and powers the system
Glove Assembly	Wearable Platform	Houses sensors and electronic components

Functional Requirements

The proposed Smart Glove incorporates multiple intelligent functions to achieve accurate lock validation and eliminate false detections.

Sensor Fusion Logic

- The system shall combine data from the MPU6050 motion sensor and PVDF vibration sensor.
- Motion information shall be used to identify operator hand movement.
- Vibration information shall be used to verify actual lock-impact events.
- Sensor fusion shall reduce false detections caused by random movements and environmental disturbances.

State Machine Logic

The system shall operate using the following states:

State 0 – Idle State

- Continuously monitor motion data.
- Remain inactive when no significant movement is detected.

State 1 – Motion Detected State

- Trigger when acceleration exceeds the predefined threshold.
- Wait for a corresponding lock-impact event.

State 2 – Impact Confirmed State

- Validate the impact using the PVDF sensor.
- Capture vibration data for analysis.

State 3 – Analysis State

- Process captured vibration signals.
- Compare results with the Golden Template.
- Generate the final decision.

Calibration Logic

- The system shall record vibration signatures from five correctly installed harness locks.
- Each vibration signal shall undergo FFT processing.
- The resulting frequency spectra shall be averaged.
- The averaged spectrum shall be stored as a Golden Template in EEPROM memory.

FFT-Based Signal Processing

- The system shall acquire 64 vibration samples after every valid impact.
- FFT analysis shall convert time-domain vibration signals into frequency-domain data.
- Frequency characteristics shall be extracted for pattern recognition.

Golden Template Matching

- The system shall compare real-time FFT output with the stored Golden Template.
- A similarity score shall be calculated based on spectral differences.
- The match percentage shall determine whether the lock installation is valid.

Real-Time Feedback System

- Green LED indication shall represent a successful lock installation.
- Red LED indication shall represent a defective or mismatched lock installation.
- Feedback shall be provided immediately after analysis.

Data Storage Function

- Calibration data shall be stored permanently in EEPROM memory.
- The stored Golden Template shall automatically load during system startup.
- Recalibration shall be possible whenever required.

False Detection Elimination

- The system shall ignore pre-impact hand movements.
- Random vibrations, accidental touches, and environmental noise shall not trigger lock validation.
- Only validated motion-impact sequences shall initiate FFT analysis.

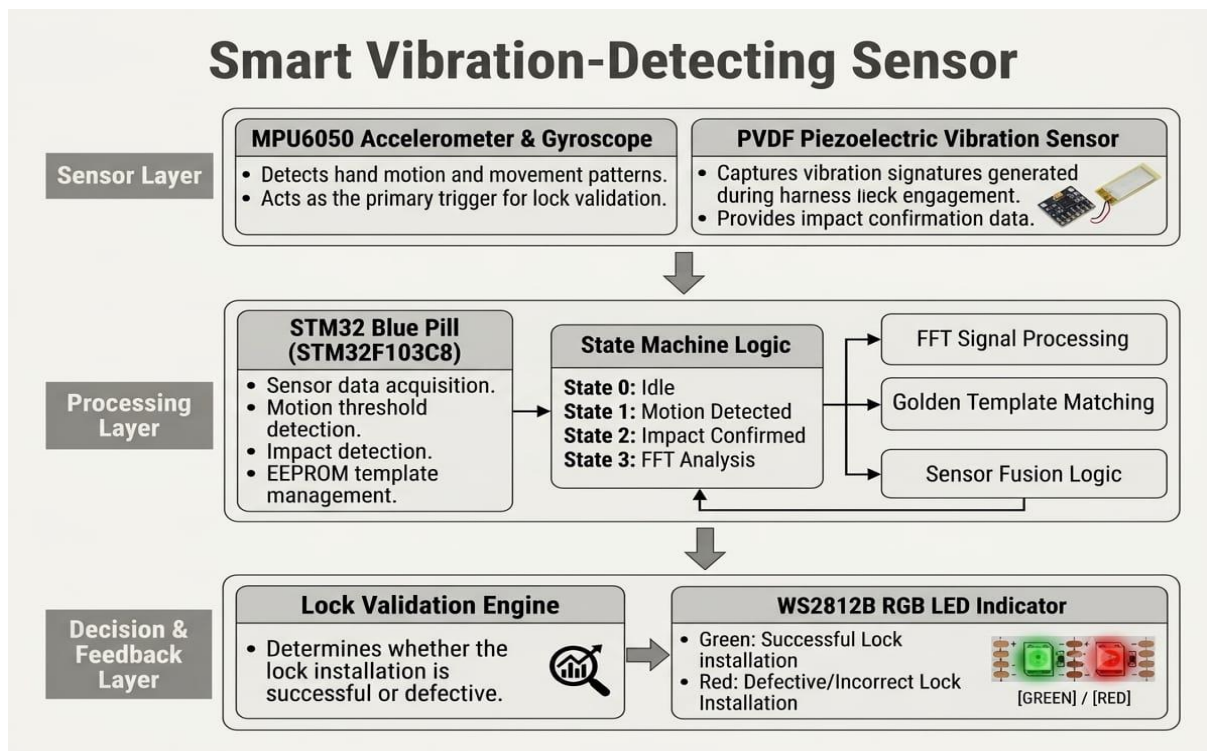
CHAPTER 4

SYSTEM ARCHITECTURE

Architecture Overview

The proposed Industrial Smart Glove follows a three-layer architecture consisting of the Sensor Layer, Processing Layer, and Decision & Feedback Layer. The system utilizes sensor fusion, state-machine logic, FFT-based vibration analysis, and template matching to accurately validate harness lock installations while minimizing false detections.

System Architecture Diagram



Sensor Layer (Input Layer)

The Sensor Layer is responsible for collecting real-time data from the assembly operation. It acts as the primary input stage of the system.

Components

MPU6050 Motion Sensor

- Detects hand movement and acceleration during lock installation.
- Identifies whether the operator is performing a locking action.
- Serves as the first trigger for the state machine.

PVDF Piezoelectric Vibration Sensor

- Captures vibration signals generated during harness lock engagement.
- Detects impact events and lock-click vibrations.
- Provides high-sensitivity vibration data for FFT analysis.

Functions

- Motion detection.
- Impact detection.
- Real-time signal acquisition.
- Input generation for sensor fusion.

Processing Layer (STM32 State Machine Logic Layer)

The Processing Layer is the core intelligence unit of the Smart Glove system. It processes sensor data, performs signal analysis, and determines lock quality.

Components

STM32 Blue Pill (STM32F103C8)

- Central controller of the system.
- Acquires sensor data.
- Executes all processing algorithms.

Sensor Fusion Logic

- Combines MPU6050 and PVDF sensor data.
- Validates that both motion and impact occur together.
- Eliminates false detections caused by random vibrations.

State Machine Logic

The system operates through four states:

State 0 – Idle

- Waits for hand movement.

State 1 – Motion Detected

- Verifies operator motion.

State 2 – Impact Confirmed

- Confirms lock impact using PVDF sensor.

State 3 – Analysis

- Performs FFT and lock validation.

Calibration Logic

- Records five successful lock installations.
- Generates a reference vibration profile.
- Creates the Golden Template.

FFT Signal Processing

- Converts vibration signals from time domain to frequency domain.
- Extracts unique lock vibration characteristics.

Golden Template Matching

- Compares live FFT spectrum with stored template.
- Calculates similarity percentage.
- Determines lock quality.

EEPROM Memory Management

- Stores calibration data permanently.
- Loads Golden Template during startup.

Functions

- Sensor fusion.

- State machine execution.
- FFT analysis.
- Template matching.
- Decision generation.

Decision & Feedback Layer (Output Layer)

The Decision & Feedback Layer communicates the final lock validation result to the operator.

Components

Lock Validation Engine

- Receives similarity score from processing layer.
- Determines whether the lock is successful or defective.

WS2812B RGB LED Indicator

- Provides immediate visual feedback.

Feedback Logic

Green LED

- Lock installation matches Golden Template.
- Indicates successful lock engagement.

Red LED

- Lock installation does not match Golden Template.
- Indicates defective or incomplete locking.

Functions

- Real-time operator feedback.
- Quality verification.
- Error indication.
- Production assistance.

CHAPTER 5

APPLICATION LOGIC AND FEATURE BREAK DOWN

5.1 Sensor Input System

The Sensor Input System serves as the primary data acquisition layer of the Smart Glove. It consists of two sensors: the MPU6050 Motion Sensor and the PVDF Piezoelectric Vibration Sensor. These sensors work together through sensor fusion to accurately identify harness lock installation events.

MPU6050 Motion Sensor

The MPU6050 is a 3-axis accelerometer and gyroscope used to detect hand movement during assembly operations. The sensor continuously monitors the operator's hand motion and acts as the first validation stage of the system. The STM32 Blue Pill reads acceleration values from the MPU6050 and compares them against a predefined motion threshold. In the implemented code, the motion threshold is set to 2.0 g-force.

```
float MOTION_THRESHOLD = 2.0;
```

When the measured acceleration exceeds this threshold, the system assumes that the operator is performing a locking action and proceeds to the next stage of validation.

PVDF Piezoelectric Vibration Sensor

The PVDF Piezoelectric Sensor is responsible for detecting vibration and impact generated during harness lock engagement. Unlike conventional sound-based systems, the PVDF sensor captures the actual mechanical vibration produced by the locking event. This enables more reliable detection and minimizes false triggers caused by environmental noise.

The impact threshold is defined as:

```
int IMPACT_THRESHOLD = 2000;
```

Only when the vibration amplitude exceeds this threshold does the system consider the event as a valid lock impact and continue with signal analysis.

Signal Processing and Filtration

To achieve reliable lock validation, the Smart Glove employs multiple filtering stages before making a decision. The objective is to eliminate false detections caused by random hand movements, accidental contacts, machine vibrations, or environmental disturbances.

Motion Threshold Filtering

The first filtration stage uses MPU6050 acceleration data. Small hand movements below the threshold value are ignored.

Condition:

```
abs(accZ) > MOTION_THRESHOLD
```

This ensures that only intentional assembly actions trigger further processing.

Impact Threshold Filtering

Once motion is detected, the system continuously monitors the PVDF sensor for a valid impact.

Condition:

```
analogRead(PIEZO_PIN) > IMPACT_THRESHOLD
```

This second filtration layer confirms that an actual locking impact has occurred rather than random movement.

FFT-Based Signal Filtering

After impact confirmation, the vibration signal is sampled and converted into a frequency spectrum using the Fast Fourier Transform (FFT). Frequency-domain analysis enables the system to identify the unique vibration characteristics of a successful lock installation while rejecting unrelated vibrations and noise.

Match Threshold Validation

The final filtering stage compares the real-time FFT spectrum with the stored Golden Template.

```
float FFT_MATCH_THRESHOLD = 85.0;
```

Only signals with a similarity score above 85% are considered successful lock installations.

State Machine Logic

The Smart Glove operates using a state-machine architecture that ensures systematic processing of locking events. This approach improves reliability and prevents unnecessary FFT computations.

State 0 – Idle State

In the Idle State, the system continuously monitors MPU6050 acceleration data while remaining inactive.

Condition:

- Acceleration ≤ 2.0 g

Action:

- Continue monitoring motion data.

State 1 – Motion Detected State

When the acceleration exceeds the motion threshold, the system transitions to the Motion Detected State.

Condition:

- Acceleration > 2.0 g

Action:

- Start monitoring the PVDF sensor.
- Wait for impact confirmation.

A timeout of 200 milliseconds is implemented to avoid remaining indefinitely in this state.

State 2 – Impact Confirmed State

Once the PVDF sensor detects an impact greater than the threshold value, the system confirms a valid locking event.

Condition:

- Piezo Value > 2000

Action:

- Capture vibration samples.
- Begin FFT analysis.

FFT Analysis and Template Comparison

The system captures 64 vibration samples immediately after impact detection.

```
#define SAMPLES 64
```

```
#define SAMPLING_FREQUENCY 1000
```

During calibration, vibration signatures from five correctly installed locks are recorded. The FFT spectrum of each lock is calculated and averaged to generate a Golden Template.

The comparison process follows:

1. Detect motion using MPU6050.
2. Confirm impact using PVDF sensor.
3. Capture 64 vibration samples.
4. Perform FFT analysis.
5. Compare FFT spectrum with Golden Template.
6. Calculate similarity percentage.
7. Validate against the 85% match threshold.

This multi-stage process forms the core Sensor Fusion and State Machine Logic of the Smart Glove.

Event Generation and Output

The STM32 Blue Pill acts as the central processing unit and generates output events based on the results of vibration analysis.

Successful Lock Installation

If the FFT similarity score is greater than or equal to 85%, the lock installation is classified as successful.

Output:

```
setLED(0,255,0);
```

Indication:

- Green LED ON
- Successful lock engagement detected

Defective Lock Installation

If the FFT similarity score falls below the threshold, the lock installation is classified as defective.

Output:

```
setLED(255,0,0);
```

Indication:

- Red LED ON
- Incorrect or incomplete lock installation

EEPROM Storage

The system stores calibration data permanently in EEPROM memory. Whenever the glove is powered ON, the stored Golden Template is automatically loaded and used for lock validation.

This eliminates the need for repeated calibration during normal operation.

Real-Time Serial Monitoring

The Arduino IDE Serial Monitor is used for debugging, calibration, and real-time system monitoring.

System Startup Messages

When the system starts, status information is displayed:

--- Smart Glove System Started ---

Golden Template Loaded from EEPROM.

or

WARNING: Not Calibrated! Type 'C' to Calibrate.

Calibration Command

The calibration process is initiated through the Serial Monitor by entering:

C

or

c

Upon receiving the command, the system enters calibration mode.

>>> Entering Calibration Mode...

Calibration Started.

Install 5 PERFECT locks now.

Waiting for impact...

Lock Detection During Calibration

As each correct lock is detected, the system records its vibration signature and displays progress messages.

Recording Good Lock #1

Recording Good Lock #2

Recording Good Lock #3

Recording Good Lock #4

Recording Good Lock #5

After recording five successful locks:

Calculating Average...

Saving to EEPROM...

Calibration Complete! System Ready.

Real-Time Validation Messages

During operation, state transitions and validation results are continuously displayed.

Examples:

State: Motion Detected

State: Impact Detected -> Running FFT

Match Score: 92%

RESULT: SUCCESS (Green)

or

Match Score: 63%

RESULT: FAILURE (Red)

These serial messages provide valuable insight into system behavior, calibration status, FFT matching results, and lock validation decisions, enabling easy debugging and performance evaluation during industrial deployment.

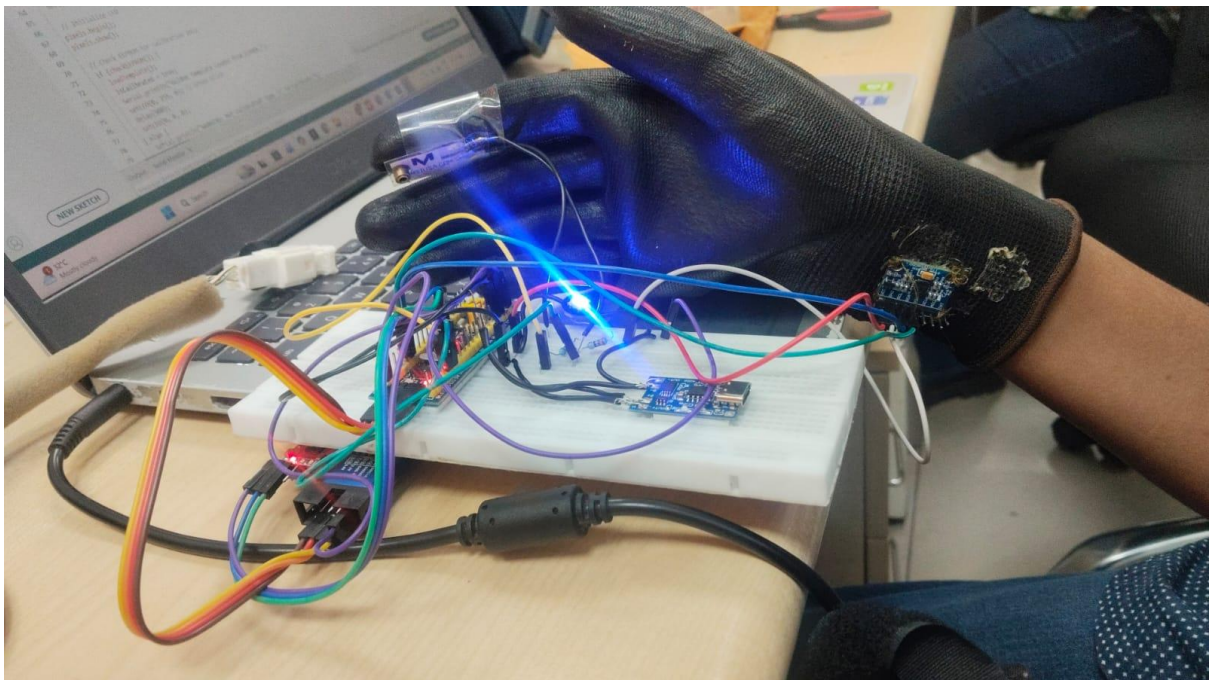
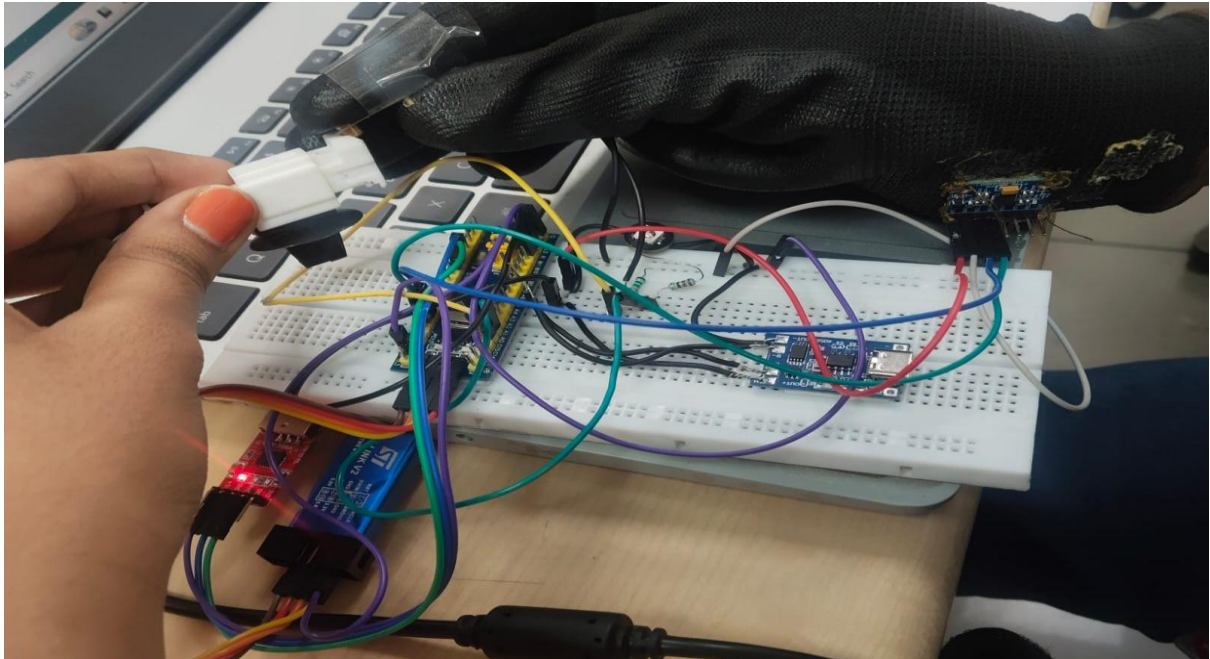
CHAPTER 6

CONCLUSION

The Industrial Smart Glove for Zero False Detection of Harness Lock Installation was successfully designed and developed to improve quality assurance in automotive assembly operations. The system integrates an MPU6050 motion sensor, a PVDF piezoelectric vibration sensor, and an STM32 Blue Pill microcontroller to accurately detect and validate harness lock installations. By utilizing sensor fusion, state machine logic, FFT-based vibration analysis, and Golden Template matching, the proposed solution effectively distinguishes genuine lock engagements from random movements and environmental disturbances. This approach significantly reduces false detections compared to conventional sound-based or manual inspection methods.

The developed system provides real-time feedback to operators through LED indications, enabling immediate identification of successful and defective lock installations. The calibration mechanism ensures reliable operation by creating a reference vibration profile from multiple successful locking events. Overall, the project demonstrates the practical implementation of embedded systems, digital signal processing, and smart wearable technology in industrial environments. The proposed solution contributes to improved assembly accuracy, reduced rework, enhanced productivity, and supports Industry 4.0 initiatives for intelligent manufacturing and quality control.

OUTCOME OF THE PROJECT



FUTURE SCOPE

Although the developed Smart Glove provides reliable harness lock validation, several enhancements can be incorporated in future versions to further improve its performance, adaptability, and industrial applicability.

1. The system can be expanded to support multiple types of locks, including plastic clips, electrical connectors, snap-fit locks, and other fastening mechanisms by creating separate Golden Templates for different applications.\
2. Machine Learning and Artificial Intelligence algorithms can be integrated to automatically learn and classify vibration signatures, improving accuracy and eliminating the need for manual threshold tuning.
3. The current prototype can be redesigned as a compact Surface Mount Device (SMD)-based PCB, resulting in a smaller, lightweight, and fully wearable industrial product.
4. Wireless communication technologies such as Bluetooth Low Energy (BLE), Wi-Fi, or LoRa can be integrated to transmit lock validation data to manufacturing dashboards, quality monitoring systems, or cloud platforms.
5. A mobile application or web-based monitoring system can be developed for real-time visualization of lock validation results, calibration status, and production analytics.
6. Additional sensors such as force sensors, pressure sensors, or acoustic sensors can be incorporated to provide multi-modal verification and further improve detection reliability.
7. Advanced adaptive calibration techniques can be implemented to automatically adjust thresholds based on operator behavior, environmental conditions, and different assembly stations.

8. Data logging and predictive analytics can be added to track installation trends, identify recurring defects, and support preventive quality maintenance in manufacturing environments.

9. Integration with Manufacturing Execution Systems (MES) and Industry 4.0 platforms can enable complete traceability of lock installation operations and support smart factory implementations.

10. Future research can focus on developing a universal smart glove capable of validating various assembly operations across multiple industries, including automotive, aerospace, electronics, and industrial equipment manufacturing.